GCSE CHEMISTRY

Higher Tier – Paper 2 style

Topic 4.6: Rate and extent of chemical change

Practice questions in the style of AQA GCSE Chemistry (8462). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 130 minutes (roughly 1 minute per mark)
Materials	For this paper you must have: a ruler, a pencil, a calculator and the Periodic Table at the back of the question section.
Instructions	• Use black ink or a black ball-point pen (pencil for diagrams and graphs). • Answer **all** questions. • Answer the questions in the spaces provided. • Do all rough work on a separate sheet. • Cross through any work you do not want to be marked.
Information	• The maximum mark for this paper is **126**. • The marks for questions are shown in brackets. • You are allowed to use a calculator. • You are reminded of the need for good English and clear presentation in your answers. • Questions that ask you to *describe* or *explain* are marked for the science you include, so give each point its own sentence.

Name: _____ Date: _____ Mark: _____ / 126

0 1 This question is about the rate of chemical reactions.

0 1 . 1 Name **three** factors that change the rate of a chemical reaction. **[3 marks]**

0 1 . 2 A student collects 48 cm^3 of hydrogen in 120 seconds. **[2 marks]**
Calculate the mean rate of reaction in cm^3/s.

Answer = _____ cm^3/s

0 1 . 3 In another reaction a flask loses 2.4 g in mass in 80 seconds. **[2 marks]**
Calculate the mean rate of reaction in g/s.

Answer = _____ g/s

0 1 . 4 0.45 mol of a product forms in 90 seconds. Calculate the mean rate in mol/s. **[2 marks]**

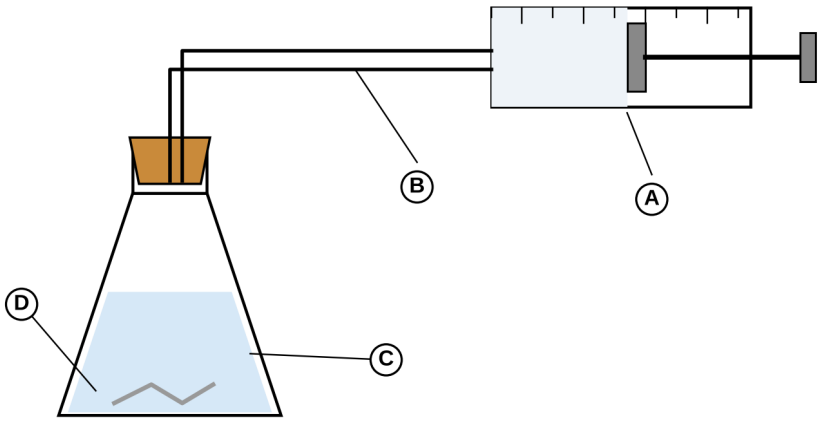
Answer = _____ mol/s

0 1 . 5 Which change would **increase** the rate of reaction of magnesium with dilute hydrochloric acid? **[1 mark]**

*Tick **one** box.*

- ☐ Using a lower temperature
- ☐ Using more concentrated acid
- ☐ Using one large piece of magnesium
- ☐ Using a smaller volume of the same acid

0 2 A student investigates the rate of reaction between magnesium ribbon and dilute hydrochloric acid. The diagram shows the apparatus.



0 2 . 1 Name the apparatus labelled **A** and **C**. **[2 marks]**

0 2 . 2 Name the gas collected in the apparatus and give a test for it. **[2 marks]**

0 2 . 3 Write a balanced symbol equation for the reaction. **[2 marks]**

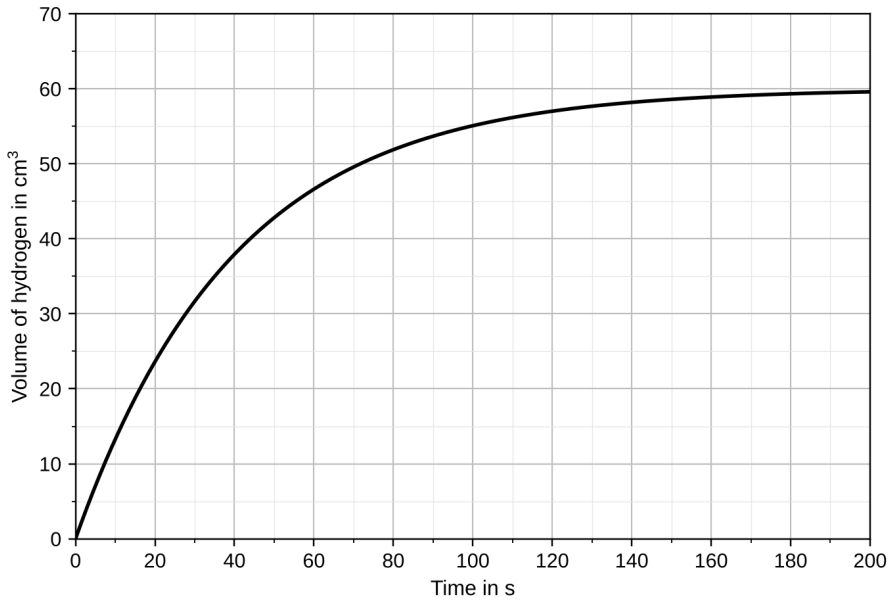
0 2 . 4 The student must put the bung in the flask quickly after adding the magnesium. Suggest why. **[1 mark]**

0 2 . 5 Describe **one** other way of measuring the rate of this reaction. **[1 mark]**

0 2 . 6 The student wants to know how the rate changes as the temperature changes. Describe **two** things he must keep the same. **[2 marks]**

0 2 . 7 Give **one** safety precaution for this experiment. **[1 mark]**

0 3 The graph shows the volume of hydrogen collected when a piece of magnesium reacts with an excess of dilute hydrochloric acid.



0 3 . 1 What is the total volume of hydrogen collected? **[1 mark]**

0 3 . 2 How long does it take to collect 30 cm^3 of hydrogen? **[1 mark]**

0 3 . 3 Calculate the mean rate of reaction for the first 40 seconds. **[3 marks]**

Answer = _____ cm^3/s

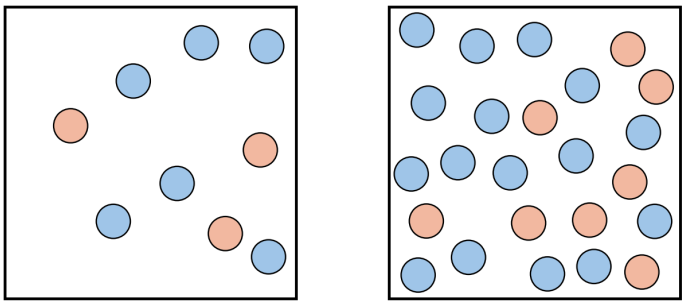
0 3 . 4 Draw a tangent to the curve at 20 seconds. Use it to calculate the rate of reaction at 20 seconds. **[3 marks]**

Answer = _____ cm^3/s

0 3 . 5 Explain why the rate of reaction decreases as the reaction goes on. **[2 marks]**

0 3 . 6 The experiment is repeated at a higher temperature. All other conditions are the same. **[2 marks]**
Sketch the curve you would expect on the grid.

0 4 The diagrams show two solutions of the same reactants. The circles represent particles. Solution **B** is more concentrated than solution **A**.



A B

0 4 . 1 Explain, using collision theory, why the reaction is faster in solution **B**. **[3 marks]**

0 4 . 2 What is meant by **activation energy**? **[1 mark]**

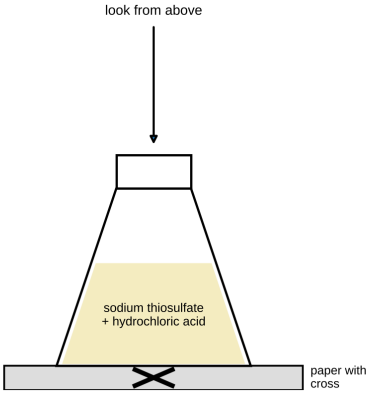
0 4 . 3 Explain, in terms of particles, why increasing the temperature increases the rate of reaction. **[3 marks]**

0 4 . 4 Magnesium powder reacts faster with acid than the same mass of magnesium ribbon. Explain why. **[2 marks]**

0 4 . 5 Explain why increasing the pressure of a gas increases the rate of a reaction between gases. **[2 marks]**

0 4 . 6 Some collisions between particles do not lead to a reaction. Give **one** reason why. **[1 mark]**

0 5 A student investigates how temperature affects the rate of the reaction between sodium thiosulfate solution and dilute hydrochloric acid. A cloudy precipitate of sulfur forms. The student times how long it takes for a cross under the flask to disappear.



0 5 . 1 Complete the equation for the reaction. **[1 mark]**

$Na_2S_2O_3 + 2HCl \rightarrow 2NaCl + ____ + SO_2 + H_2O$

0 5 . 2 Why does the cross **disappear**? **[1 mark]**

0 5 . 3 Sulfur dioxide is a toxic gas. How should the student reduce the risk? **[1 mark]**

0 5 . 4 Give **two** variables the student must control. **[2 marks]**

0 5 . 5 The results are shown. Complete the table by calculating the rate for each temperature. **[2 marks]**

Rate = 1000 ÷ time

Temperature in °C	Time for cross to disappear in s	Rate = 1000 ÷ time
20	150	6.7
30	82	12.2
40	46	
50	25	
60	15	

0 5 . 6 Describe the pattern shown by the results. **[2 marks]**

0 5 . 7 Different people may disagree about when the cross has disappeared. Suggest a better method of finding the end point. **[1 mark]**

0 6 A cube of calcium carbonate has sides of 2 cm. The cube is cut into eight smaller cubes with sides of 1 cm.

0 6 . 1 Calculate the surface area to volume ratio of the 2 cm cube. Give your answer in the form **x : 1**. **[3 marks]**

Answer = _____ : 1

0 6 . 2 Calculate the surface area to volume ratio of one 1 cm cube. **[2 marks]**

Answer = _____ : 1

0 6 . 3 The small cubes react with acid faster than the large cube. Use your answers to explain why. **[2 marks]**

0 7 Hydrogen peroxide solution slowly decomposes to form water and oxygen. Manganese(IV) oxide is added as a catalyst and the reaction speeds up.

0 7 . 1 Write a balanced symbol equation for the decomposition of hydrogen peroxide (H_2O_2). **[2 marks]**

0 7 . 2 Why does manganese(IV) oxide not appear in the equation? **[1 mark]**

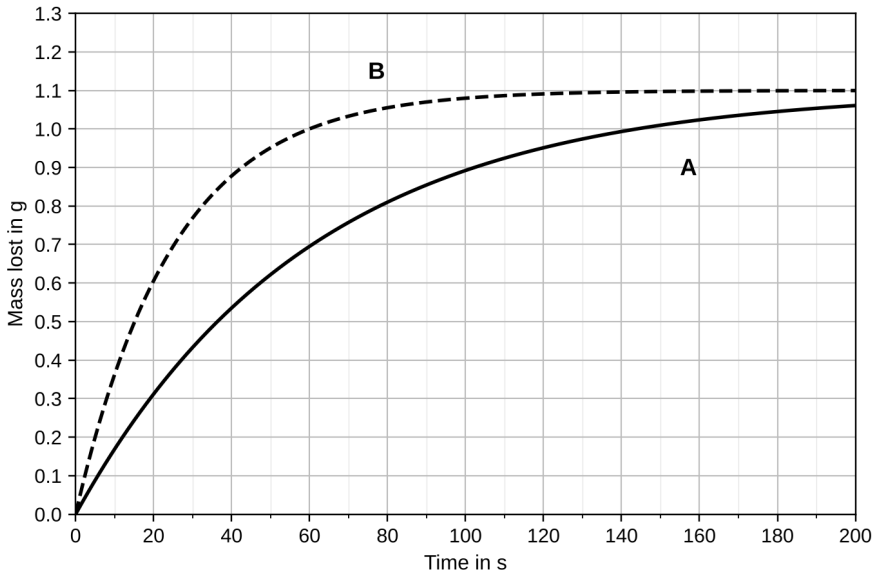
0 7 . 3 Explain how a catalyst increases the rate of reaction. **[2 marks]**

0 7 . 4 Name the catalyst used in the Haber process. **[1 mark]**

0 7 . 5 Give **two** reasons why catalysts are used in industrial processes. **[2 marks]**

0 7 . 6 Industrial catalysts are often used as pellets or fine gauzes. Suggest why. **[1 mark]**

0 8 A student reacts 2.5 g of calcium carbonate (marble chips) with excess dilute hydrochloric acid in a flask on a balance. A piece of cotton wool is placed in the neck of the flask. The graph shows the loss in mass for large chips (**A**) and small chips (**B**).



0 8 . 1 Which curve, **A** or **B**, shows the small chips? Give a reason. **[2 marks]**

0 8 . 2 Explain why the mass of the flask decreases. **[2 marks]**

0 8 . 3 Why is cotton wool placed in the neck of the flask? **[1 mark]**

0 8 . 4 Calculate the mean rate of reaction for curve **B** between 0 and 50 seconds. **[3 marks]**

Answer = _____ g/s

0 8 . 5 Both reactions finish with the same total mass lost. Explain why. **[2 marks]**

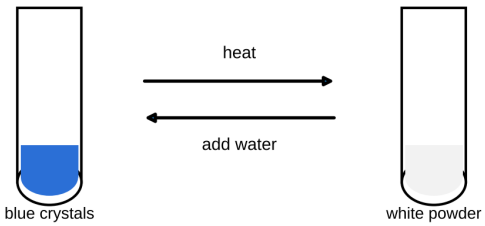
0 8 . 6 The total mass lost is 1.1 g. Calculate the mass of calcium carbonate that reacted. **[3 marks]**

$CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$

(Relative formula masses: $CO_2 = 44$, $CaCO_3 = 100$)

Answer = _____ g

0 9 Blue hydrated copper(II) sulfate crystals change to white anhydrous copper(II) sulfate when heated. The change can be reversed by adding water.



0 9 . 1 What does the symbol ⇌ show about a reaction? **[1 mark]**

0 9 . 2 Write the equation for this reaction using the symbol ⇌. **[2 marks]**

Hydrated copper(II) sulfate is $CuSO_4.5H_2O$.

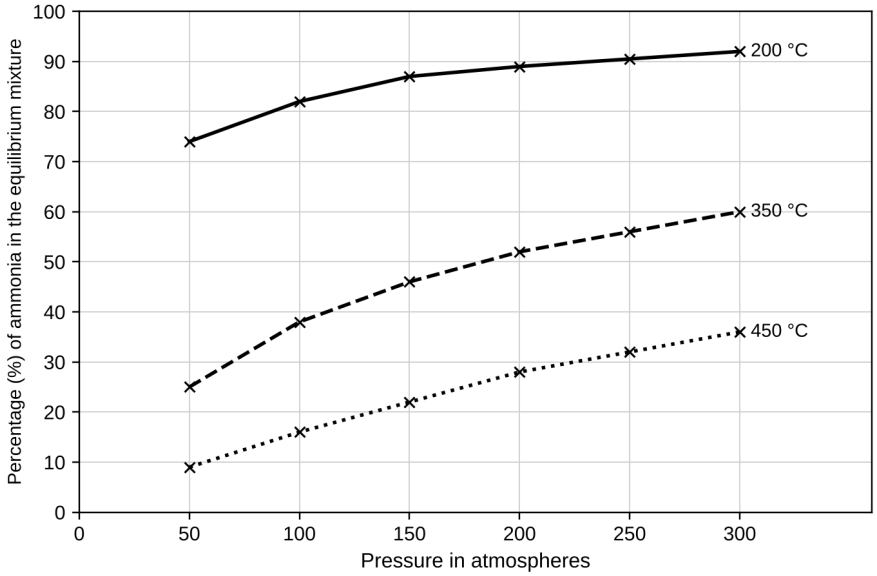
0 9 . 3 Describe what happens when water is added to white anhydrous copper(II) sulfate. **[2 marks]**

0 9 . 4 The forward reaction is endothermic. What does this tell you about the reverse reaction? **[2 marks]**

0 9 . 5 Suggest how this reaction can be used as a chemical test for water. **[1 mark]**

0 9 . 6 Ammonium chloride decomposes when heated: $NH_4Cl \rightleftharpoons NH_3 + HCl$ **[2 marks]**
A student heats ammonium chloride at the bottom of a test tube. White solid appears again near the top. Explain why.

1 0 Ammonia is made from nitrogen and hydrogen in the Haber process.
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ The forward reaction is exothermic.
The graph shows how the percentage of ammonia in the equilibrium mixture changes with pressure and temperature.



1 0 . 1 What is meant by **dynamic equilibrium**? **[2 marks]**

1 0 . 2 Why must the reaction be carried out in a closed system for equilibrium to be reached? **[1 mark]**

1 0 . 3 Use the graph to find the percentage of ammonia in the equilibrium mixture at 200 atmospheres and 450 °C. **[1 mark]**

1 0 . 4 What is the effect of increasing the temperature on the percentage of ammonia at equilibrium? Use the graph and explain your answer. **[3 marks]**

10.5 Explain why increasing the pressure increases the percentage of ammonia at equilibrium. **[2 marks]**

10.6 The Haber process uses a temperature of 450 °C, even though a lower temperature gives a higher percentage of ammonia. Explain why 450 °C is used. **[2 marks]**

10.7 Adding an iron catalyst does not change the percentage of ammonia at equilibrium. Explain why. **[2 marks]**

11 Dinitrogen tetroxide (colourless) is in equilibrium with nitrogen dioxide (brown) in a sealed tube.
$N_2O_4(g) \rightleftharpoons 2NO_2(g)$ The forward reaction is endothermic.

11.1 Complete the table. For each change, state whether the equilibrium moves to the left, moves to the right or does not change, and say what happens to the colour. **[4 marks]**

Change	Position of equilibrium	Colour of the gas mixture
increase the temperature		
increase the pressure		
decrease the pressure		
add a catalyst		

11.2 State the effect on the position of equilibrium if some nitrogen dioxide is removed from the tube. **[1 mark]**

1 1 . 3 State Le Chatelier's principle. **[2 marks]**

1 2 Stretch questions (grade 7–9).

1 2 . 1 *Explain, using collision theory, how changing the temperature, concentration, surface area and use of a catalyst each affect the rate of a reaction. **[6 marks]**

1 2 . 2 *Chemical companies use a compromise temperature and pressure in the Haber process. **[6 marks]**

Explain why, using ideas about the rate of reaction and the position of equilibrium.

END OF QUESTIONS

The Periodic Table of the Elements (elements 1–56)

1 **H** 1																	4 **He** 2
7 **Li** 3	9 **Be** 4											11 **B** 5	12 **C** 6	14 **N** 7	16 **O** 8	19 **F** 9	20 **Ne** 10
23 **Na** 11	24 **Mg** 12											27 **Al** 13	28 **Si** 14	31 **P** 15	32 **S** 16	35.5 **Cl** 17	40 **Ar** 18
39 **K** 19	40 **Ca** 20	45 **Sc** 21	48 **Ti** 22	51 **V** 23	52 **Cr** 24	55 **Mn** 25	56 **Fe** 26	59 **Co** 27	59 **Ni** 28	63.5 **Cu** 29	65 **Zn** 30	70 **Ga** 31	73 **Ge** 32	75 **As** 33	79 **Se** 34	80 **Br** 35	84 **Kr** 36
85.5 **Rb** 37	88 **Sr** 38	89 **Y** 39	91 **Zr** 40	93 **Nb** 41	96 **Mo** 42	[98] **Tc** 43	101 **Ru** 44	103 **Rh** 45	106 **Pd** 46	108 **Ag** 47	112 **Cd** 48	115 **In** 49	119 **Sn** 50	122 **Sb** 51	128 **Te** 52	127 **I** 53	131 **Xe** 54
133 **Cs** 55	137 **Ba** 56	57–71															

Group 1 → 18: left to right. Periods: top to bottom.

Relative atomic mass is shown above each symbol and atomic (proton) number below. Elements 57–71 are not shown.

MARK SCHEME – Topic 4.6: Rate and extent of chemical change

How to use this mark scheme: mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *science*, not the wording.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	temperature ; concentration (of solutions) / pressure (of gases) ; surface area (of solids) / catalyst ;	Any three from: temperature, concentration, pressure, surface area, catalyst.	3	AO1 4.6.1.1
01.2	48 ÷ 120 ; 0.4 (cm^3/s) ;	Mean rate = quantity of product formed ÷ time taken.	2	AO2 4.6.1.2
01.3	2.4 ÷ 80 ; 0.03 (g/s) ;		2	AO2 4.6.1.2
01.4	0.45 ÷ 90 ; 0.005 (mol/s) / 5×10^{-3} ;		2	AO2 4.6.1.2
01.5	Using more concentrated acid	Tick 'Using more concentrated acid'. Volume alone does not change concentration.	1	AO2 4.6.1.1
02.1	A: gas syringe ; C: conical flask ;	Do not accept 'syringe' alone or 'beaker'.	2	AO1 4.6.1.2
02.2	hydrogen ; lit splint gives a squeaky pop ;		2	AO1 4.4.1.1
02.3	$Mg + 2HCl \rightarrow MgCl_2 + H_2$; correct formulae and balanced ;		2	AO2 4.4.1.1
02.4	so that no gas escapes / all the gas is collected		1	AO3 4.6.1.2
02.5	measure the volume of gas by displacement of water (upturned measuring cylinder / burette) ; measure the loss in mass of the flask on a balance ;	Allow 'time how long the magnesium takes to disappear'. Any one.	1	AO1 4.6.1.2
02.6	volume and concentration of acid ; mass / length of magnesium (and same surface area / form of magnesium) ;	Any two sensible control variables.	2	AO3 4.6.1.2
02.7	wear eye protection / goggles ;	Allow 'tie back long hair'. Do not accept 'wear a lab coat' alone.	1	AO1 4.6.1.2
03.1	59 – 60 cm^3		1	AO3 4.6.1.2
03.2	26 – 30 s	Allow 28 s.	1	AO3 4.6.1.2
03.3	volume at 40 s = 37 – 38 cm^3 ; 37.9 ÷ 40 ; 0.9 – 1.0 (cm^3/s) ;	Allow ecf from the volume read. Correct answer with no working scores 3.	3	AO2 4.6.1.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.4	tangent drawn touching the curve at 20 s without crossing it ; gradient = change in volume ÷ change in time (triangle shown) ; 0.8 – 1.0 (cm^3/s) ;	The tangent method is Higher Tier only. True value is about 0.9 cm^3/s.	3	AO2 4.6.1.2
03.5	the concentration of the acid decreases / reactants are used up ; so fewer (successful) collisions per second ;		2	AO3 4.6.1.3
03.6	curve is steeper at the start ; levels off at the same final volume (60 cm^3) ;	Same amount of magnesium so the same total volume of gas.	2	AO3 4.6.1.3
04.1	more particles in the same volume ; so the particles collide more often / frequency of collisions increases ; so more successful collisions per second ;		3	AO2 4.6.1.3
04.2	the minimum energy that particles must have to react / for a collision to be successful		1	AO1 4.6.1.3
04.3	particles move faster / have more energy ; collide more frequently ; a greater proportion of collisions have energy ≥ the activation energy / more successful collisions ;	Higher Tier answers need both frequency and energy of collisions.	3	AO2 4.6.1.3
04.4	powder has a larger surface area ; so more particles are exposed / more frequent collisions ;		2	AO2 4.6.1.3
04.5	particles are closer together / more particles in the same volume ; so more frequent collisions ;		2	AO2 4.6.1.3
04.6	the particles do not have enough energy (activation energy) ; particles do not collide with the correct orientation ;	Any one.	1	AO1 4.6.1.3
05.1	S / sulfur		1	AO1 4.6.1.2
05.2	the sulfur precipitate makes the solution cloudy / opaque		1	AO2 4.6.1.2
05.3	do the experiment in a fume cupboard / well-ventilated room / use dilute solutions		1	AO1 4.6.1.2
05.4	volume (and concentration) of sodium thiosulfate solution ; volume and concentration of acid / same flask and same cross ;	Any two.	2	AO3 4.6.1.2
05.5	40 °C: 21.7 ; 50 °C: 40.0 ; 60 °C: 66.7 ; (all three correct for 2 marks; two correct for 1 mark) ;		2	AO2 4.6.1.2
05.6	as the temperature increases the rate increases ; the rate increases by a factor of about 1.8 for every 10 °C rise / rate nearly doubles ;		2	AO3 4.6.1.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.7	use a light sensor / data logger to measure the transmitted light		1	AO3 4.6.1.2
06.1	surface area = 6 × 2 × 2 = 24 cm^2 ; volume = 2 × 2 × 2 = 8 cm^3 ; 24 ÷ 8 = 3 : 1 ;		3	AO2 4.6.1.1
06.2	surface area = 6 × 1 × 1 = 6 cm^2 and volume = 1 cm^3 ; 6 : 1 ;		2	AO2 4.6.1.1
06.3	the small cubes have a larger surface area to volume ratio (6 : 1 compared with 3 : 1) ; more particles exposed to the acid so more frequent collisions ;		2	AO3 4.6.1.3
07.1	$2H_2O_2 \rightarrow 2H_2O + O_2$; correct formulae and balanced ;		2	AO2 4.6.1.4
07.2	it is not used up / is not a reactant or product / is unchanged at the end		1	AO2 4.6.1.4
07.3	provides an alternative reaction pathway with a lower activation energy ; more particles have enough energy to react so more successful collisions per second ;		2	AO1 4.6.1.4
07.4	iron		1	AO1 4.6.1.4
07.5	reactions are faster so more product made in a given time ; the reaction can be done at a lower temperature, saving energy / costs / fuel ;	Allow 'catalysts can be reused'. Any two.	2	AO2 4.6.1.4
07.6	increases surface area (so catalysis is faster)		1	AO3 4.6.1.4
08.1	B ; the curve is steeper / the reaction is faster (larger surface area) ;		2	AO3 4.6.1.3
08.2	carbon dioxide is made ; and escapes from the flask (as a gas) ;		2	AO2 4.3.1.1
08.3	to stop acid spray escaping (but let gas out)		1	AO3 4.6.1.2
08.4	mass lost at 50 s = 0.94 – 0.96 g ; 0.95 ÷ 50 ; 0.018 – 0.020 g/s ;		3	AO2 4.6.1.2
08.5	same mass of calcium carbonate (the limiting reactant) ; so the same mass of carbon dioxide is made ;		2	AO3 4.6.1.3
08.6	moles CO_2 = 1.1 ÷ 44 = 0.025 ; moles $CaCO_3$ = 0.025 (1 : 1 ratio) ; mass = 0.025 × 100 = 2.5 g ;		3	AO2 4.3.2.1
09.1	the reaction is reversible / can go in both directions		1	AO1 4.6.2.1
09.2	$CuSO_4.5H_2O \rightleftharpoons CuSO_4 + 5H_2O$; correct formulae and the reversible arrow ;		2	AO1 4.6.2.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
09.3	it turns blue ; it gets hot (exothermic) ;		2	AO2 4.6.2.1
09.4	the reverse reaction is exothermic ; the same amount of energy is transferred (released) as was taken in ;		2	AO2 4.6.2.2
09.5	anhydrous copper(II) sulfate turns from white to blue if water is present		1	AO3 4.6.2.1
09.6	ammonia and hydrogen chloride gases rise and cool ; the reverse reaction happens, re-forming ammonium chloride (solid) ;		2	AO3 4.6.2.1
10.1	the forward and reverse reactions occur at the same rate ; so the concentrations of reactants and products stay constant (reactions have not stopped) ;		2	AO1 4.6.2.3
10.2	so no reactants or products can escape		1	AO2 4.6.2.3
10.3	27 – 29 %		1	AO3 4.6.2.3
10.4	percentage of ammonia decreases ; the forward reaction is exothermic ; so the equilibrium moves in the endothermic direction (to the left) when temperature increases ;	Le Chatelier: a system at equilibrium opposes any change in conditions.	3	AO3 4.6.2.4
10.5	4 moles of gas on the left and 2 moles of gas on the right ; equilibrium moves to the side with fewer gas molecules (right) ;		2	AO3 4.6.2.4
10.6	at lower temperatures the rate of reaction is too slow ; 450 °C is a compromise between a reasonable rate and a reasonable yield ;		2	AO3 4.6.2.4
10.7	a catalyst speeds up the forward and reverse reactions equally ; so equilibrium is reached faster but the position does not change ;		2	AO3 4.6.2.3
11.1	increase temperature: right ; darker brown ; increase pressure: left ; paler brown / lighter ; decrease pressure: right ; darker brown ; add a catalyst: no change ; no change in colour ; (1 mark for each correct row) ;	Increasing pressure favours the side with fewer gas molecules (left, 1 mole). Catalyst: no change in position.	4	AO3 4.6.2.4
11.2	moves to the right (to make more nitrogen dioxide)		1	AO2 4.6.2.4
11.3	if a change is made to the conditions of a system at equilibrium ; the system responds to counteract / oppose the change ;		2	AO1 4.6.2.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
12.1	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • rate depends on the frequency of successful collisions between particles • increasing temperature: particles move faster so collide more often and more of the collisions have at least the activation energy • increasing concentration (or pressure): more particles in the same volume so more frequent collisions • increasing surface area of a solid: more particles exposed so more frequent collisions • a catalyst gives an alternative pathway with a lower activation energy so more collisions are successful • a catalyst is not used up and does not change the products or the overall energy change		6	AO1 4.6.1.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
12.2	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • the forward reaction (making ammonia) is exothermic so a low temperature would give a higher yield • but at a low temperature the rate is too slow / equilibrium takes too long to reach, so 450 °C is a compromise • increasing the pressure moves the equilibrium to the right (fewer gas molecules) and increases the rate • but very high pressure needs strong, expensive equipment and a lot of energy, and is more dangerous, so about 200 atmospheres is used • an iron catalyst speeds up the reaction without changing the equilibrium position, allowing a lower temperature to be used • ammonia is liquefied and removed so the equilibrium keeps shifting right; unreacted nitrogen and hydrogen are recycled		6	AO3 4.6.2.4

Where marks are lost in Topic 4.6

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Mean rate** = quantity of product (or reactant used) ÷ time. Write the unit (cm^3/s, g/s or mol/s). For a tangent: draw it so it touches the curve, make a big triangle, then change in y ÷ change in x.
- ☐ **Collision theory answers need two ideas:** particles collide *more often* (frequency) and/or *more of the collisions are successful* (energy ≥ activation energy). Temperature affects both; concentration, pressure and surface area only change frequency.
- ☐ **Never say** 'more collisions' on its own. Say 'more *frequent* collisions' or 'more *successful* collisions per second'.
- ☐ **Reading graphs:** steeper = faster. A curve that levels off has finished. The same final volume or mass means the same amount of limiting reactant.
- ☐ **Catalysts:** lower the activation energy by giving an alternative pathway; not used up; do not change the equilibrium position or the products.
- ☐ **Dynamic equilibrium:** forward and reverse reactions continue at the *same rate*; concentrations are constant; needs a closed system. Do not say the reactions have 'stopped'.
- ☐ **Le Chatelier:** the system opposes the change. Pressure ↑ → side with fewer gas molecules. Temperature ↑ → endothermic direction. Concentration ↑ of a substance → moves away from it.
- ☐ **Compromise conditions:** link every condition to *both* yield and rate (and cost/safety). A lower temperature gives more product but takes too long.
- ☐ **Required practical:** disappearing cross uses 1 ÷ time as the rate. Control the volumes and concentrations, and say why a precipitate hides the cross.